

Ibn Tofail University

Analysis V

Problem Set I

Exercise 1:

Show that:

$$|x + y| \leq |x| + |y|, \quad \forall x, y \in \mathbb{R}^n$$

Correction

$$\begin{aligned} \|a + b\|^2 &= \langle a + b, a + b \rangle \\ &= \langle a, a \rangle + 2\langle a, b \rangle + \langle b, b \rangle \\ &= \|a\|^2 + 2\langle a, b \rangle + \|b\|^2 \\ &\leq \|a\|^2 + 2\|a\|\|b\| + \|b\|^2 \\ &= (\|a\| + \|b\|)^2 \\ \implies \|a + b\| &\leq \|a\| + \|b\| \end{aligned}$$

ive used C.S to prove it. (C.S : $|xy| \leq |x||y|$)

Exercise 2:

Let $\|\cdot\|$ be the real function defined on $\mathbb{R}^2$ by:

$$\|u = (x, y)\| = \sqrt{x^2 + y^2}.$$

Show that this function satisfies:

$$\|u + v\| \leq \|u\| + \|v\|, \quad \forall u, v \in \mathbb{R}^2$$

Correction

This is just the same as exercise 1, but in $\mathbb{R}^2$

Exercise 3:

Let $X = (x_1, \dots, x_n) \in \mathbb{R}^n$. We consider the following applications:

- The N_1 -norm (or Manhattan norm):

$$N_1(X) = \sum_{i=1}^n |x_i|$$

- The N_2 -norm (or Euclidean norm):

$$N_2(X) = \sqrt{\sum_{i=1}^n |x_i|^2}$$

- The N_∞ -norm (or maximum norm):

$$N_\infty(X) = \sup_{i=1, \dots, n} |x_i|$$

Show that the three preceding applications are norms on $\mathbb{R}^n$.

Correction

1. The Three Norms verifies the first condition all of them are positive and when $X = 0 \iff N(X) = 0$. (it's obvious)
2. The Three Norms verifies the second condition all of them are absolutely scalable. (again it's obvious)
3. The Three Norms verifies the third condition (Triangle inequality) all of them satisfy: $N(X + Y) \leq N(X) + N(Y)$.

For N_1 and N_∞ it's obvious, for N_2 we can use C.S inequality to prove it.

Exercise 4:

Show that the preceding norms satisfy the following properties:

1. $N_\infty(X) \leq N_1(X) \leq nN_\infty(X)$
2. $N_\infty(X) \leq N_2(X) \leq \sqrt{n}N_\infty(X)$
3. $\frac{1}{n}N_1(X) \leq N_2(X) \leq \sqrt{n}N_1(X)$

Correction

1. Since $N_\infty(X) = |x_{i_0}|$ for some i_0 ,

$$N_1(X) = \sum_{i=1}^n |x_i| \geq |x_{i_0}| = N_\infty(X),$$

and since $|x_i| \leq N_\infty(X)$ for all i ,

$$N_1(X) = \sum_{i=1}^n |x_i| \leq \sum_{i=1}^n N_\infty(X) = n N_\infty(X).$$

2. Similarly,

$$N_2(X)^2 = \sum_{i=1}^n |x_i|^2 \geq |x_{i_0}|^2 = N_\infty(X)^2 \implies N_2(X) \geq N_\infty(X),$$

and

$$N_2(X)^2 = \sum_{i=1}^n |x_i|^2 \leq \sum_{i=1}^n N_\infty(X)^2 = n N_\infty(X)^2 \implies N_2(X) \leq \sqrt{n} N_\infty(X).$$

3. By the Cauchy–Schwarz inequality,

$$\left(\sum_{i=1}^n |x_i| \cdot 1 \right)^2 \leq \left(\sum_{i=1}^n |x_i|^2 \right) \left(\sum_{i=1}^n 1^2 \right) = n N_2(X)^2,$$

so

$$N_1(X)^2 \leq n N_2(X)^2 \implies N_2(X) \geq \frac{1}{\sqrt{n}} N_1(X) \geq \frac{1}{n} N_1(X).$$

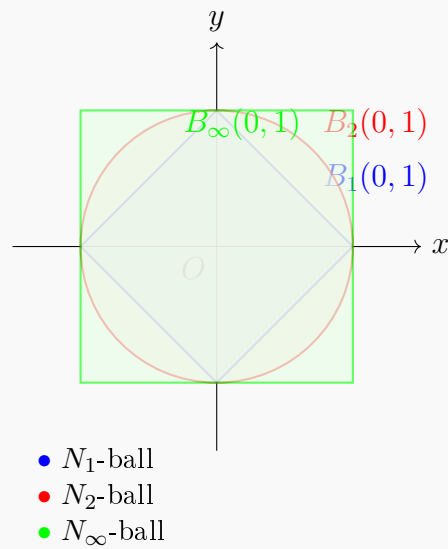
Also, since $|x_i|^2 \leq |x_i| \cdot N_1(X)$ for each i , $N_2(X)^2 = \sum_{i=1}^n |x_i|^2 \leq N_1(X) \sum_{i=1}^n |x_i| = N_1(X)^2$

$$\implies N_2(X) \leq N_1(X) \leq \sqrt{n} N_1(X)$$

————— EXERCISE 5 —————

Exercise 5:

In $\mathbb{R}^2$, draw the open balls of center O and radius 1, for each of the three usual norms.

Correction**Explanation:**

1. The N_1 -ball is the set $\{(x, y) : |x| + |y| < 1\}$.
2. The N_2 -ball is the set $\{(x, y) : x^2 + y^2 < 1\}$.
3. The N_∞ -ball is the set $\{(x, y) : \max(|x|, |y|) < 1\}$.

Exercise 6:

1. Show that an open ball is an open set.
2. Show that a closed ball is a closed set.

Correction

Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$ or $\mathbb{C}$. For any $a \in E$ and $r > 0$, define:

1. Open ball: $B(a, r) = \{x \in E : \|x - a\| < r\}$
2. Closed ball: $\overline{B}(a, r) = \{x \in E : \|x - a\| \leq r\}$

1. An open ball is an open set.

Let $x_0 \in B(a, r)$. We must show there exists $\varepsilon > 0$ such that $B(x_0, \varepsilon) \subseteq B(a, r)$.

Set $\varepsilon = r - \|x_0 - a\| > 0$ (since $x_0 \in B(a, r)$).

Take any $x \in B(x_0, \varepsilon)$. Then:

$$\|x - a\| \leq \|x - x_0\| + \|x_0 - a\| < \varepsilon + \|x_0 - a\| = r.$$

So $x \in B(a, r)$, hence $B(x_0, \varepsilon) \subseteq B(a, r)$.

Therefore, every point in $B(a, r)$ has a neighborhood contained in it $\rightarrow B(a, r)$ is open.

2. A closed ball is a closed set.

We show its complement is open. Let $x_0 \notin \overline{B}(a, r)$, i.e., $\|x_0 - a\| > r$.

Set $\varepsilon = \|x_0 - a\| - r > 0$.

Take any $x \in B(x_0, \varepsilon)$. Then:

$$\|x - a\| \geq \|x_0 - a\| - \|x - x_0\| > \|x_0 - a\| - \varepsilon = r.$$

So $x \notin \overline{B}(a, r)$, hence $B(x_0, \varepsilon) \subseteq E \setminus \overline{B}(a, r)$.

Thus, the complement of $\overline{B}(a, r)$ is open $\rightarrow \overline{B}(a, r)$ is closed.